



**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIORSERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2023

**MATHEMATICS P2/WISKUNDE V2
MARKING GUIDELINE/NASIENRIGLYN**

MARKS/PUNTE: 150

This marking guideline consists of 13 pages.
Hierdie nasienriglyn bestaan uit 13 bladsye.

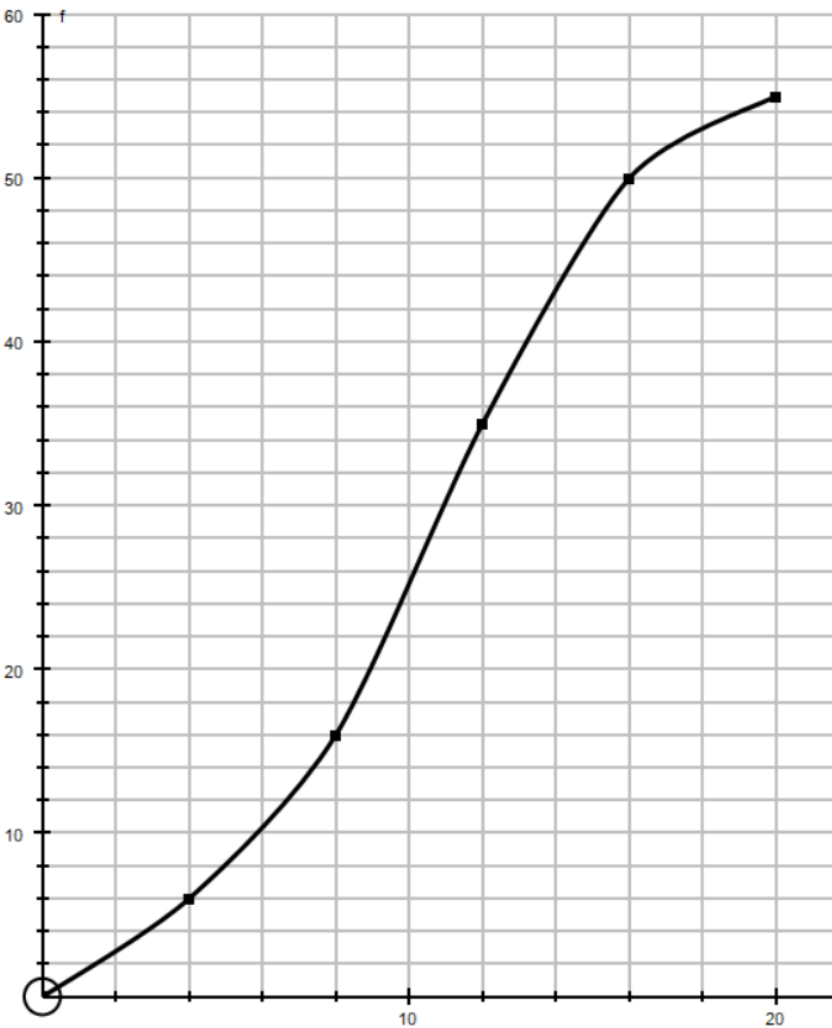
QUESTION 1/VRAAG 1



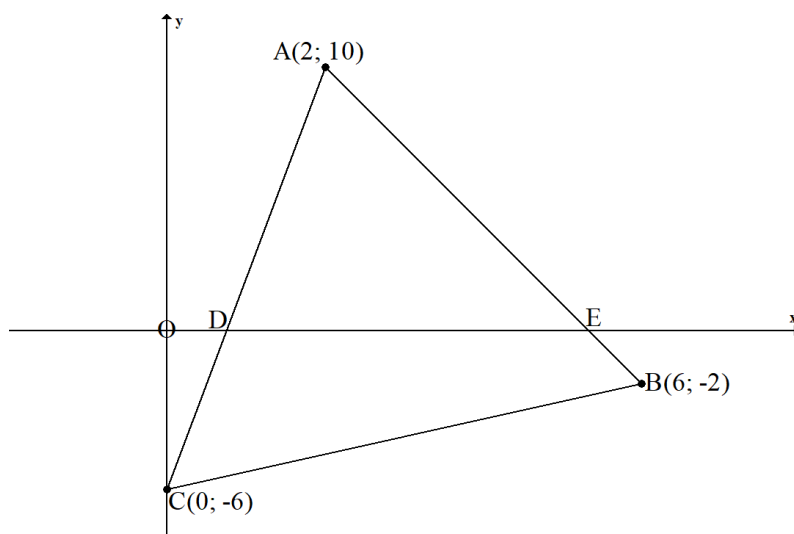
1.1	6 boys / <i>seuns</i>	✓ answer / <i>antwoord</i>	(1)
1.2	Skewed to the right / positively skewed <i>Skeef na regs / Positief skeef</i>	✓ answer / <i>antwoord</i>	(1)
1.3	$127 - 68 = 59$	✓ working out / <i>bewerking</i> ✓ answer / <i>antwoord</i>	(2)
1.4.1	Median/ <i>Mediaan</i> = 84 / Remains unchanged / <i>Bly onveranderd</i>	✓ answer / <i>antwoord</i>	(1)
1.4.2	$(87 \times 24 + 20) \div 24 = 87,8$ OR/OF 88	✓ working out / <i>bewerking</i> ✓ answer / <i>antwoord</i>	(2)
			[7]

QUESTION 2/VRAAG 2

2.1	<table><tr><th>Time waiting / <i>Wagtyd</i> (minutes / <i>minute</i>)</th><th>Number of learners <i>Aantal leerders</i></th><th>Cumulative frequency <i>Kumulatiewe frekwensie</i></th></tr><tr><td>$0 < x \leq 4$</td><td>6</td><td>6</td></tr><tr><td>$4 < x \leq 8$</td><td>10</td><td>16</td></tr><tr><td>$8 < x \leq 12$</td><td>19</td><td>35</td></tr><tr><td>$12 < x \leq 16$</td><td>15</td><td>50</td></tr><tr><td>$16 < x \leq 20$</td><td>5</td><td>55</td></tr></table>	Time waiting / <i>Wagtyd</i> (minutes / <i>minute</i>)	Number of learners <i>Aantal leerders</i>	Cumulative frequency <i>Kumulatiewe frekwensie</i>	$0 < x \leq 4$	6	6	$4 < x \leq 8$	10	16	$8 < x \leq 12$	19	35	$12 < x \leq 16$	15	50	$16 < x \leq 20$	5	55	✓ for / vir 6; 16; 35 ✓ for / vir 50; 55	(2)
	Time waiting / <i>Wagtyd</i> (minutes / <i>minute</i>)	Number of learners <i>Aantal leerders</i>	Cumulative frequency <i>Kumulatiewe frekwensie</i>																		
	$0 < x \leq 4$	6	6																		
	$4 < x \leq 8$	10	16																		
	$8 < x \leq 12$	19	35																		
	$12 < x \leq 16$	15	50																		
	$16 < x \leq 20$	5	55																		

2.2		✓ for grounding vir anker ✓ for endpoint vir eindpunt ✓✓ for any 2 correct points vir enige 2 korrekte punte	(4)
2.3	$55 - 43 = 12$ learners / <i>leerders</i> (accept/aanvaar 11 en 13)	✓ 55 ✓ 43 ✓ 12	(3)
2.4	$8 < x \leq 12$	✓ answer / antwoord	(1)
2.5	$IQR / IKW = 14 - 7 = 7$	✓ UQ ✓ LQ ✓ answer / antwoord	(3)
			[13]

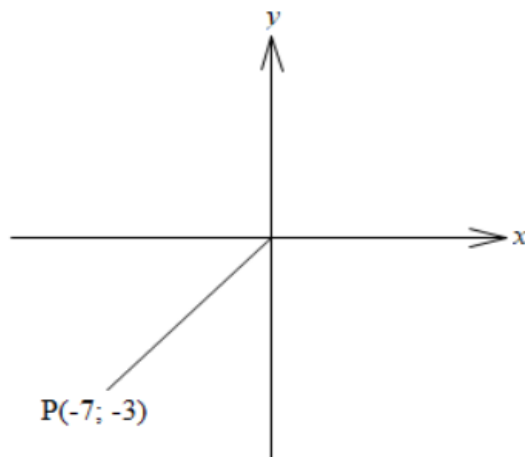
QUESTION 3 / VRAAG 3

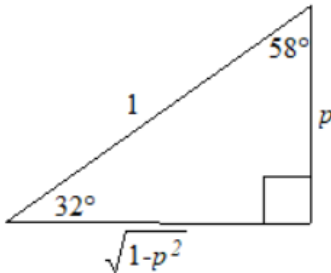


3.1	$m_{AB} = m_{BC}$ $\frac{-8-0}{x-5+8} = \frac{-14-0}{x+8}$ $\frac{-8}{x+3} = \frac{-14}{x+8}$ $-8x - 64 = -14x - 42$ $6x = 22$ $x = \frac{11}{3}$	$\checkmark \frac{-8-0}{x-5+8}$ $\checkmark \frac{-14-0}{x+8}$ $\checkmark -8x - 64$ $\checkmark -14x - 42$ $\checkmark x = \frac{11}{3}$	(5)
3.2.1	$M\left(\frac{6+0}{2}; \frac{-2-6}{2}\right)$ $= M(3; -4)$	$\checkmark \text{substitution / vervanging}$ $\checkmark \text{answer / antwoord}$	(2)
3.2.2	$AC = \sqrt{(2-0)^2 + (10-(-6))^2}$ $AC = \sqrt{4+256}$ $AC = 2\sqrt{65}$	$\checkmark \text{substitution / vervanging}$ $\checkmark \text{answer / antwoord}$	(2)
3.2.3	$F(-4; 6)$	$\checkmark x \quad \checkmark y$	(2)
3.2.4	$m_{AB} = \frac{10-(-2)}{2-6} = \frac{12}{-4} = -3$ $m_{BC} = \frac{-2+6}{6-0} = \frac{4}{6} = \frac{2}{3}$ $m_{AB} \times m_{BC} \neq -1$ $\therefore \triangle ABCF \text{ is not a rectangle / is nie 'n reghoek nie}$ OR/OF $AC = \sqrt{260} = 2\sqrt{65}$ $AB = \sqrt{160} = 4\sqrt{10}$ $BC = \sqrt{50} = 5\sqrt{2}$ $260 \neq 160 + 50$ $AC^2 \neq AB^2 + BC^2$	$\checkmark m_{AB}$ $\checkmark m_{BC}$ $\checkmark m_{AB} \times m_{BC} \neq -1$ $\checkmark AB = \sqrt{160} = 4\sqrt{10}$ $\checkmark BC = \sqrt{50} = 5\sqrt{2}$ $\checkmark 260 \neq 160 + 50$	(3)

3.2.5	$m_{AB} = -3$ $\therefore y = -3x + c$ $10 = -3(2) + c$ $c = 16$ $\therefore y = -3x + 16$ $m_{AC} = \frac{10 - (-6)}{2 - 0} = 8$ $\therefore y = 8x - 6$	✓ m_{AB} ✓ substitution / <i>vervang</i> ✓ equation of AB / <i>vergelyking van AB</i> ✓ gradient of AC / <i>gradiënt van AC</i> ✓ equation of / <i>vergelyking van AC</i>	(5)
3.2.6	$\tan \widehat{AEX} = -3$ $\therefore \widehat{AEX} = 108,43^\circ$ $\tan \widehat{ADE} = 8$ $\therefore \widehat{ADE} = 82,87^\circ$ $\widehat{A} = 108,43^\circ - 82,87^\circ = 25,56^\circ$	✓ $\tan \widehat{AEX} = -3$ ✓ $\widehat{AEX}$ ✓ $\tan \widehat{ADE} = 8$ ✓ $\widehat{ADE}$ ✓✓ answer / <i>antwoord</i>	(6)
3.2.7	$D\left(\frac{3}{4}; 0\right)$ $E\left(\frac{16}{3}; 0\right)$ $\therefore DE = \frac{55}{12}$ units / <i>eenhede</i> Perpendicular height / <i>Loodregte hoogte</i> = 10 units / <i>eenhede</i> $\therefore A = \frac{1}{2} \times \frac{55}{12} \times 10 = 22,92 \text{ units}^2 / \text{eenhede}^2$ OR/OF $AD = 10,08$ $AE = 10,54$ $\text{Area of } \triangle ADE = \frac{1}{2} AD \cdot AE \sin A$ $\text{Area of } \triangle ADE = \frac{1}{2} (10,08)(10,54) \sin 25,56$ $\text{Area of } \triangle ADE = 22,92 \text{ units/eenhede}^2$	✓ coordinates of D / <i>koördinate van D</i> ✓ coordinates of E / <i>koördinate van E</i> ✓ length of DE / <i>lengte van DE</i> ✓ height / <i>hoogte</i> ✓ answer / <i>antwoord</i> ✓ $AD = 10,08$ ✓ $AE = 10,54$ ✓ Formula/formule ✓ substitution/vervang ✓ answer/antwoord	(5)
			[30]

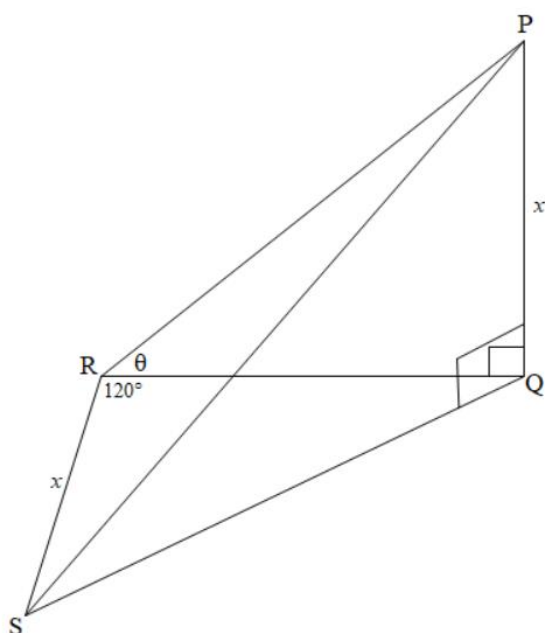
QUESTION 4 / VRAAG 4



4.1	$\tan \theta = \frac{3}{7}$ $r^2 = (-7)^2 + (-3)^2$ $r^2 = 58$ $r = \sqrt{58}$ $\frac{\sin \theta + \cos \theta}{\frac{2 \sin \theta}{\frac{-3}{\sqrt{58}} + \left(\frac{-7}{\sqrt{58}}\right)}}$ $= \frac{-10}{\sqrt{58}} \times \frac{\sqrt{58}}{-6}$ $= \frac{5}{3}$		✓ correct quadrant / <i>korrekte kwadrant</i> ✓ $\tan \theta = \frac{3}{7}$ ✓ for/vir ✓ substitution / <i>vervanging</i> ✓ simplification / <i>vereenvoudiging</i> ✓ answer / <i>antwoord</i>		(6)
					
4.2.1	$\tan(-32^\circ)$ $= -\tan 32^\circ$ $= -\frac{p}{\sqrt{1-p^2}}$		✓ method used / triangle <i>metode gebruik / driehoek</i> ✓ $-\tan 32^\circ$ ✓ answer / <i>antwoord</i>		(3)
4.2.2	$\sin 418^\circ$ $= \sin 58^\circ$ $= \sqrt{1-p^2}$	$= \sin(90^\circ - 32^\circ)$ $= \cos 32^\circ$ $= \sqrt{1-p^2}$	✓ $\sin 58^\circ$ ✓ answer / <i>antwoord</i>	✓ $\cos 32^\circ$ ✓ answer <i>antwoord</i>	(2)

4.3	$\frac{\sin 120^\circ \cdot \tan 300^\circ}{\cos(-60^\circ) \cdot \tan 225^\circ}$ $= \frac{\sin 60^\circ \cdot -\tan 60^\circ}{\cos 60^\circ \cdot \tan 45^\circ}$ $= \frac{\frac{\sqrt{3}}{2} \cdot -\frac{\sqrt{3}}{1}}{\frac{1}{2} \cdot 1}$ $= -\frac{3}{2} \times \frac{2}{1} = -3$	✓ $\sin 60^\circ$ ✓ $-\tan 60^\circ$ ✓ $\cos 60^\circ$ ✓ $\tan 45^\circ$ ✓ for numerator S/A / <i>vir teller</i> ✓ for denominator S/A / <i>vir noemer</i> ✓ answer / <i>antwoord</i>	(7)
4.4	$\frac{\cos^2 x - \cos x - \sin^2 x}{2 \sin x \cos x + \sin x} = \frac{1}{\tan x} - \frac{1}{\sin x}$ $\text{LHS/LK} = \frac{\cos^2 x - \cos x - (1 - \cos^2 x)}{2 \sin x \cos x + \sin x}$ $\text{LHS} = \frac{2 \cos^2 x - \cos x - 1}{2 \sin x \cos x + \sin x}$ $\text{LHS} = \frac{(2 \cos x + 1)(\cos x - 1)}{\sin x (2 \cos x + 1)}$ $\text{LHS} = \frac{\cos x - 1}{\sin x}$ $\text{LHS} = \frac{\cos x}{\sin x} - \frac{1}{\sin x}$ $\text{LHS} = \frac{1}{\tan x} - \frac{1}{\sin x} = \text{RHS/RK}$	✓ $1 - \cos^2 x$ ✓ simplifying / <i>vereenvoudiging</i> ✓ factoring the top / <i>faktorisering teller</i> ✓ factoring the bottom / <i>faktorisering noemer</i> ✓ $\frac{\cos x}{\sin x} - \frac{1}{\sin x}$	(5)
4.5	$\sin(5\theta) = \cos(\theta - 40^\circ)$ $\sin(5\theta) = \sin[90^\circ - (\theta - 40^\circ)]$ $\sin(5\theta) = \sin(130^\circ - \theta)$ $5\theta = 130^\circ - \theta + 360^\circ \cdot k$ $6\theta = 130^\circ + 360^\circ \cdot k$ $\theta = 21,67^\circ + 60^\circ \cdot k$ OR / OF $5\theta = 180^\circ - (130^\circ - \theta) + 360^\circ \cdot k$ $4\theta = 50^\circ + 360^\circ \cdot k$ $\theta = 12,5^\circ + 90^\circ \cdot k \text{ where/waar } k \in \mathbb{Z}$ OR / OF $0 - 5\theta = 360 - (\theta - 40) + 360k$ $-4\theta = 310 + 360k$ $\theta = -77,5 - 90k$	✓ co – function / <i>ko-funksie</i> ✓ $5\theta = 130^\circ - \theta + 360^\circ \cdot k$ ✓ $\theta = 21,67^\circ + 60^\circ \cdot k$ ✓ $5\theta = 180^\circ - (130^\circ - \theta) + 360^\circ \cdot k$ ✓ $4\theta = 50^\circ + 360^\circ \cdot k$ ✓ $\theta = 12,5^\circ + 90^\circ \cdot k$ ✓ co-function / <i>ko-funksie</i> ✓ $-6\theta = -130 + 360k$ ✓ $\theta = 21,67 - 60k$ ✓ $90 - 5\theta = 360 - (\theta - 40) + 360k$ ✓ $-4\theta = 310 + 360k$ ✓ $\theta = -77,5 - 90k$	(6)
			[29]

QUESTION 5/VRAAG 5

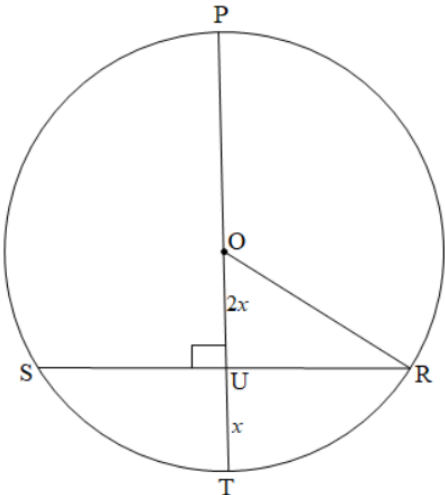
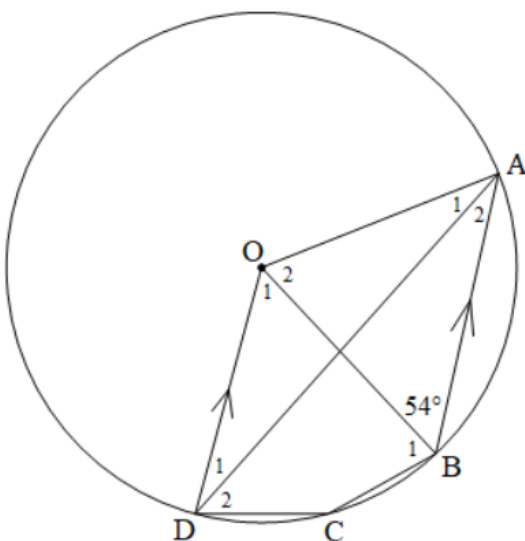


5.1	$\tan \theta = \frac{x}{QR}$ $QR = \frac{x}{\tan \theta}$	✓ $\tan \theta = \frac{x}{QR}$ ✓ answer / antwoord	(2)
5.2	$QS^2 = x^2 + \left(\frac{x}{\tan \theta}\right)^2 - 2x\left(\frac{x}{\tan \theta}\right) \cos 120^\circ$ $QS^2 = x^2 + \frac{x^2}{\tan^2 \theta} - \frac{2x^2}{\tan \theta} \left(-\frac{1}{2}\right)$ $QS^2 = x^2 + \frac{x^2}{\tan^2 \theta} + \frac{x^2}{\tan \theta}$ $QS = \sqrt{x^2 \left(1 + \frac{1}{\tan^2 \theta} + \frac{1}{\tan \theta}\right)}$ $QS = x \cdot \sqrt{\frac{1}{\tan^2 \theta} + \frac{1}{\tan \theta} + 1}$	✓ for the cosine rule vir die cosinusreël ✓ $-\frac{1}{2}$ ✓ simplification vereenvoudiging ✓ for square rooting vir vierkantswortel	(4)
5.3.1	$QS = 15 \sqrt{\frac{1}{\tan^2 22^\circ} + \frac{1}{\tan 22^\circ} + 1}$ $QS = 46,48 \text{ m}$	✓ substitution / vervanging ✓ answer / antwoord	(2)
5.3.2	$\tan Q\hat{P}S = \frac{46,48}{15}$ $Q\hat{P}S = 72,11^\circ$	✓ for using tan / vir gebruik van tan ✓ answer / antwoord	(2)
			[10]

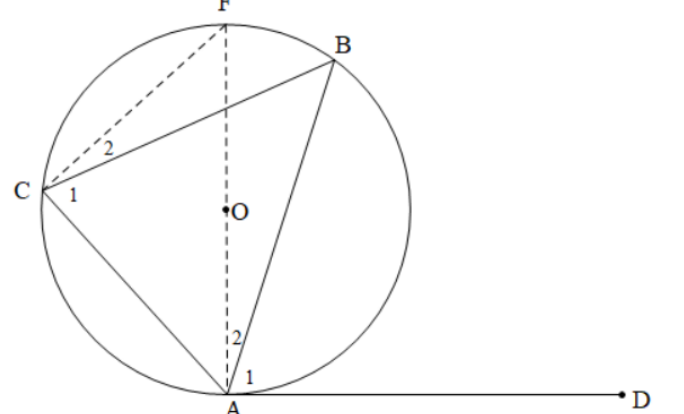
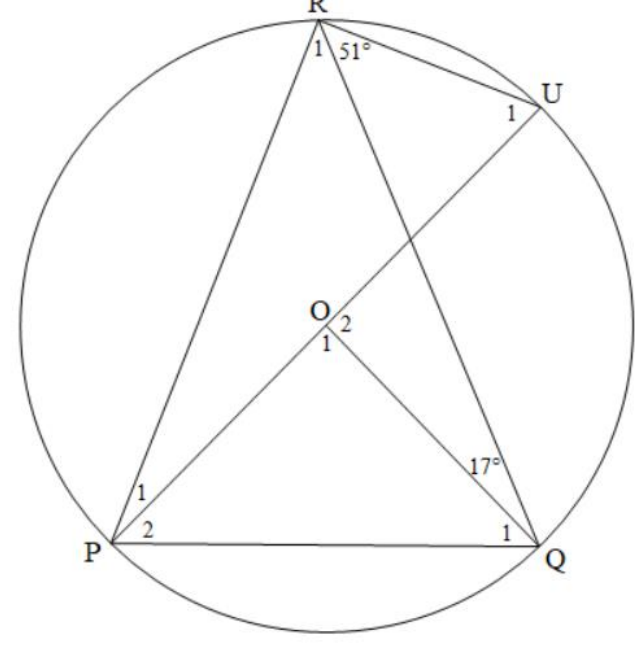
QUESTION 6 / VRAAG 6

6.1				
	For f: ✓ shape ✓ endpoints ✓ TP and/en x-intercepts Vir f: ✓ vorm ✓ eindpunte ✓ DP and/en x-afsnitte For g: ✓ asymptote ✓ shape ✓ y-intercept Vir g: ✓ asimptoot ✓ vorm ✓ y-afsnit			(6)
6.2.1	$0^\circ < x < 90^\circ$	✓ answer / antwoord		(1)
6.2.2	$-45^\circ \leq x \leq 0^\circ$ or/of $45^\circ \leq x < 90^\circ$ or/of $90^\circ < x \leq 180^\circ$	✓✓✓ for each answer vir elke antwoord		(3)
6.3	$1 \leq y \leq 3$ or/of $y \in [1; 3]$	✓✓ for the answer vir die antwoord		(2)
				[12]

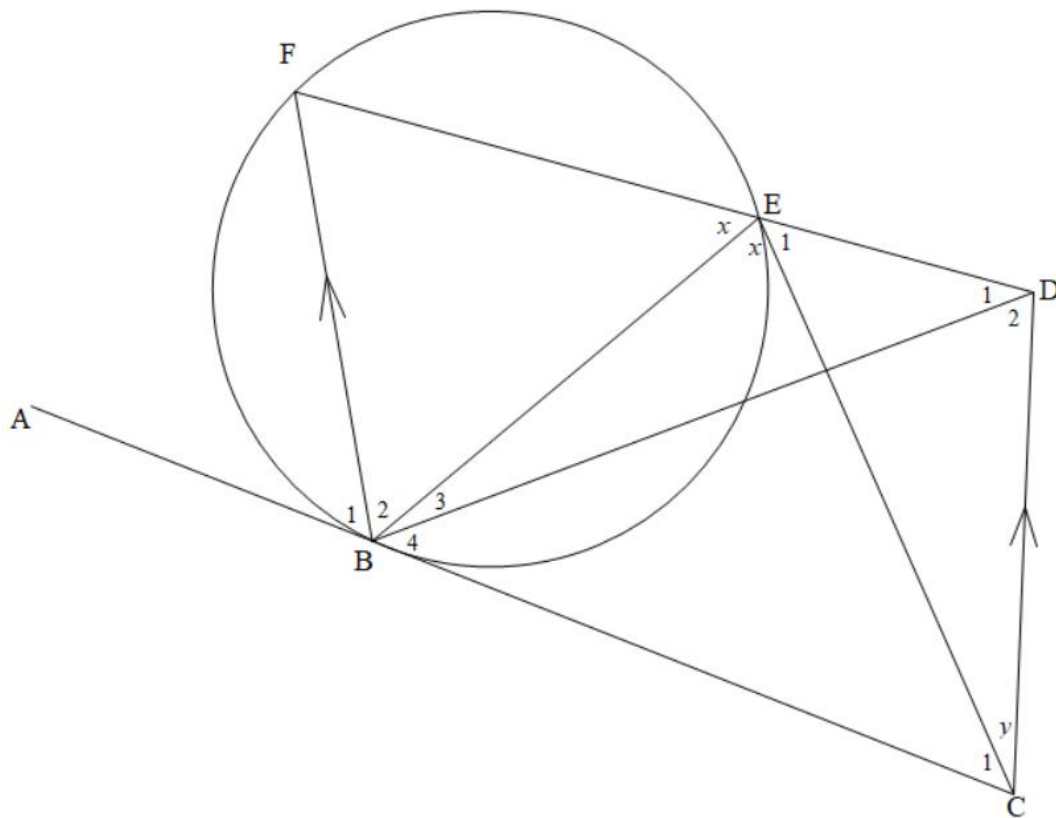
QUESTION 7 / VRAAG 7

7.1				
7.1.1	$OR = 3x$	✓ answer / antwoord	(1)	
7.1.2	$UR = 20 \text{ cm}$ (line from centre perpendicular...) (lyn vanaf middelpunt loodreg ...) $(3x)^2 = (2x)^2 + 20^2$ (Pythagoras) $9x^2 = 4x^2 + 400$ $5x^2 = 400$ $x^2 = 80$ $x = \sqrt{80} = 4\sqrt{5}$	✓ S ✓ R ✓ S and/en R ✓ simplification vereenvoudiging ✓ answer / antwoord	(5)	
7.2				
7.2.1	$\hat{O}_1 = 54^\circ$ (alt $\angle$ s / verw. $\angle$ e, $OD \parallel AB$) $\hat{A}_2 = 27^\circ$ ($\angle$ at centre) / (middelpunts $\angle$)	✓ S ✓ R ✓ S ✓ R	(4)	
7.2.2	$\hat{C} = 153^\circ$ (opp. $\angle$ s of a cq) / (teenoorst. $\angle$ e van k.v)	✓ S ✓ R	(2)	
7.2.3	$\hat{A} = 54^\circ$ ($\angle$ s opp = sides) / ($\angle$ e teenoor = sye) $\hat{A}_2 = 27^\circ$ (proven) / (reeds bewys) $\therefore \hat{A}_1 = 27^\circ$ $\therefore DA$ bisects $O\hat{A}B$ / DA halveer $O\hat{A}B$	✓ S and/en R ✓ S ✓ S	(3)	
				[15]

QUESTION 8 / VRAAG 8

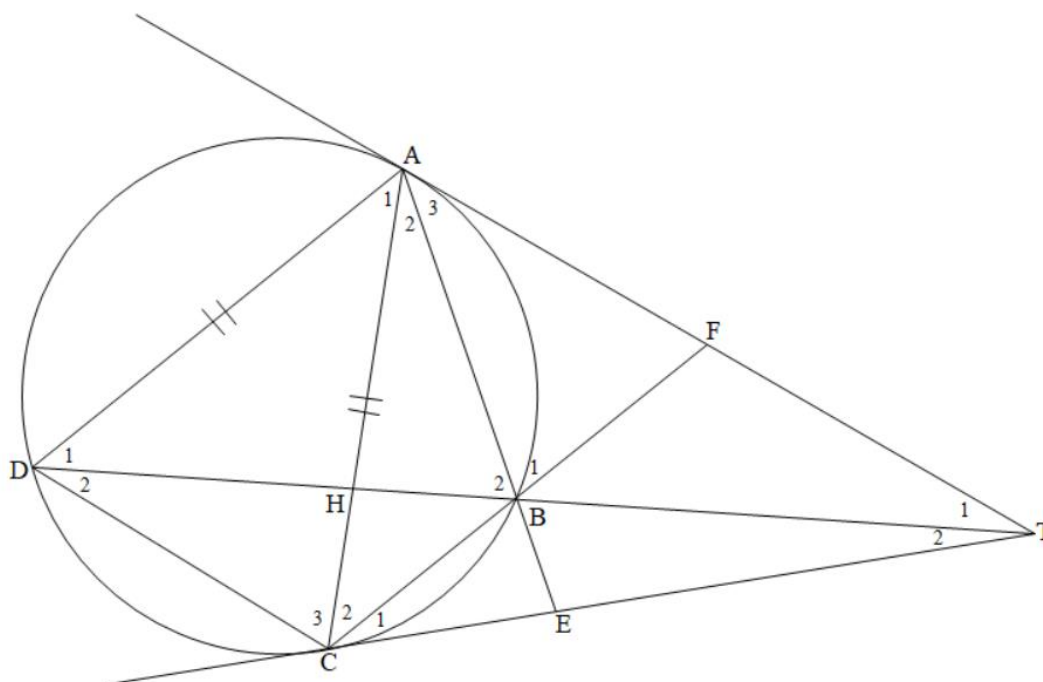
8.1	 Let /Laat $\hat{A}_1 = x$ $\therefore \hat{A}_2 = 90^\circ - x$ (tan $\perp$ rad) / (raaklyn $\perp$ radius) $\widehat{FCA} = 90^\circ$ ($\angle$ in a semi-circle) / ($\angle$ in semi-sirkel) $\hat{C}_2 = 90^\circ - x$ ($\angle$s in the same segment) / ($\angle$e in dies. segment) $\therefore \hat{C}_1 = x$ $\therefore \widehat{DAB} = \widehat{BCA}$	✓ S ✓ R ✓ S and/en R ✓ S and/en R ✓ S	(5)
8.2			
8.2.1	$\widehat{R}_1 = 39^\circ$ ($\angle$ subtended by diameter) / ($\angle$ onderspan deur middellyn)	✓ S ✓ R	(2)
8.2.2	$\widehat{O}_1 = 78^\circ$ ($\angle$ at centre) / (Middelpunts $\angle$)	✓ S ✓ R	(2)
8.2.3	$\widehat{Q}_1 = 51^\circ$ ($\angle$s opp = sides) / ($\angle$e teenoor gelyke sye)	✓ S ✓ R	(2)
8.2.4	$\widehat{U}_1 = 68^\circ$ ($\angle$s in the same segment) / ($\angle$e in dieselfde segment)	✓ S ✓ R	(2)
			[13]

QUESTION 9 / VRAAG 9



9.1.1	$\widehat{ABF} = x$ (tan-chord theorem) / (raaklyn-koord stelling)	✓ S ✓ R	(2)
9.1.2	$\widehat{BCD} = x$ (corresponding angles, $BF \parallel CD$) (ooreenkomstige hoeke ; $BF \parallel CD$)	✓ S ✓ R	(2)
9.2	Exterior angle = interior opp. angle <i>Buitehoek = teenoorstaande binnehoek.</i>	✓ R	(1)
9.3	$\widehat{BDC} = x$ (angles in the same segment) / (hoeke in dieselfde segment)	✓ S ✓ R	(4)
	$\widehat{FBD} = x$ (alternate angles $BF \parallel CD$) (verwissellende hoeke $BF \parallel CD$)	✓ S ✓ R	
			[9]

QUESTION 10 / VRAAG 10



10.1	$\widehat{D} = \widehat{C}_3$ ($\angle$ s opp. = sides) / ($\angle$ e teenoor = sye) $\widehat{D} = \widehat{B}_1$ (ext. $\angle$ of a cq) / (<i>buite $\angle$ van k.v</i>) $\widehat{C}_3 = \widehat{B}_2$ ($\angle$ s in the same segment) / ($\angle$ e in dieselfde segment) $\widehat{B}_1 = \widehat{B}_2$	$\checkmark$ S and/en R $\checkmark$ S $\checkmark$ R $\checkmark$ S and/en R	(4)
10.2	$\widehat{C}_1 + \widehat{C}_2 = \widehat{D}$ (tan-chord theorem) / (<i>raaklyn-koord stelling</i>) But/Maar $\widehat{D} = \widehat{B}_2$ (proven) / (<i>reeds bewys</i>) $\therefore \widehat{B}_2 = \widehat{C}_1 + \widehat{C}_2$ (ext $\angle$ = int opp.angle) / <i>(buite $\angle$ = teenoorst. binnehoek)</i> BECH is a c.q. / BECH is <i>n</i> kv (converse. ext. $\angle$ of a cq) / <i>(omgekeerde buite $\angle$ van kv)</i>	$\checkmark$ S and/en R $\checkmark$ S and/en R $\checkmark$ R	(3)
10.3	$\widehat{B}_2 = \widehat{A}_3 + \widehat{T}_1$ (ext $\angle$ of a triangle) / (<i>buite $\angle$ van driehoek</i>) But/Maar $\widehat{B}_2 = \widehat{C}_1 + \widehat{C}_2$ (proven) / (<i>reeds bewys</i>) And/ En AT = CT (tangents from the same point) <i>(raaklyne vanaf dieselfde punt)</i> $\therefore \widehat{C}_1 + \widehat{C}_2 = \widehat{A}_3 + \widehat{A}_2$ ($\angle$ s opp. = sides) / ($\angle$ e teenoor = sye) $\widehat{A}_3 + \widehat{A}_2 = \widehat{A}_3 + \widehat{T}_1$ (conv. tan-chord) / <i>(omgekeerde raaklyn-koord)</i> $\therefore \widehat{A}_2 = \widehat{T}_1$ CA is a tangent to circle ABT / CA is <i>n</i> raaklyn aan sirkel ABT)	$\checkmark$ S and/en R $\checkmark$ S and/en R $\checkmark$ S and/en R $\checkmark$ S and/en R $\checkmark$ S and/en R	(5)
			[12]

TOTAL/TOTAAL: 150